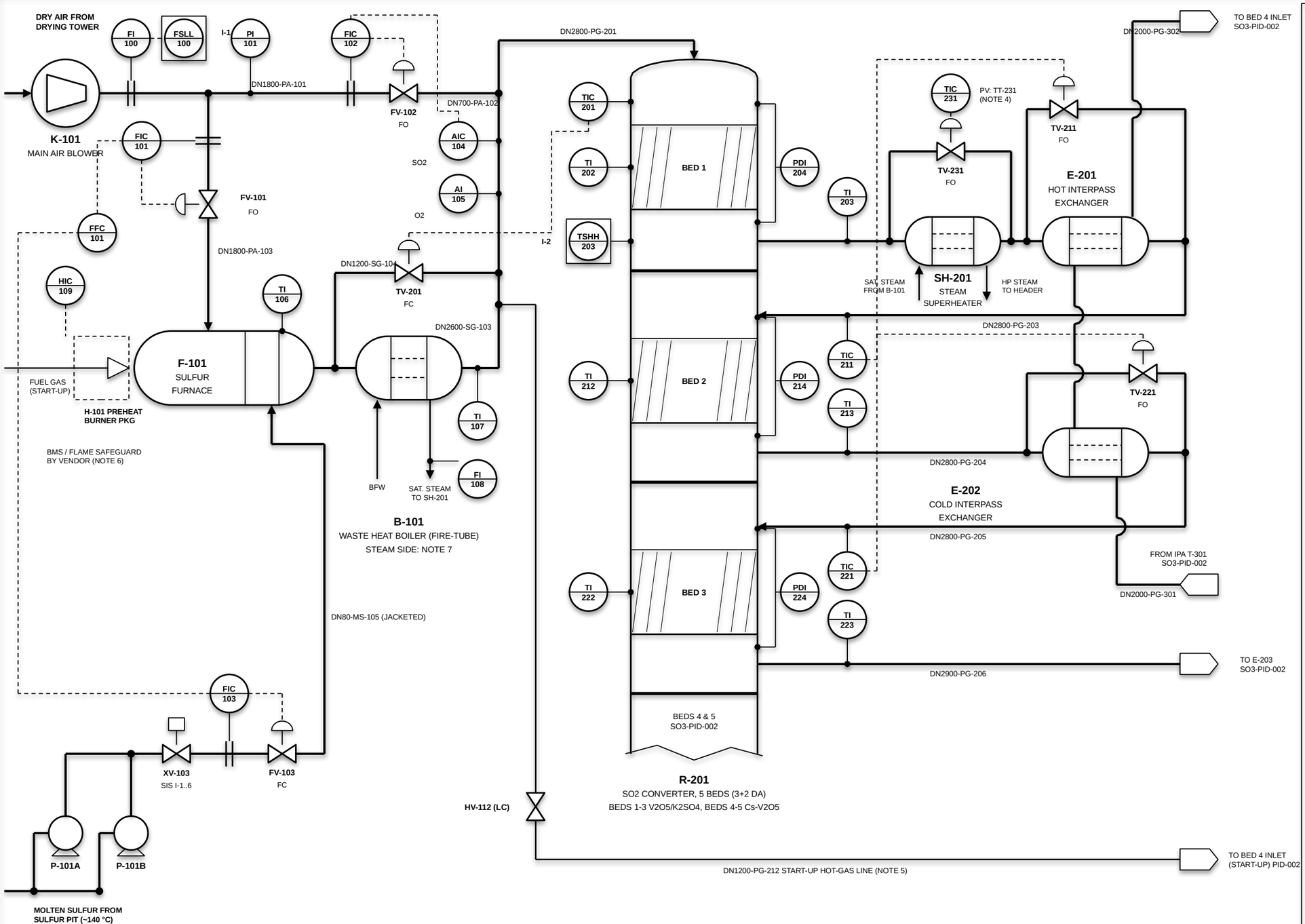


LEGEND

	PROCESS LINE		UTILITY LINE
	SIGNAL LINE		LINE JUMP (NO JOIN)
	CONTROL VALVE FO/FC		SIS SHUT-OFF VALVE
	MANUAL VALVE (LC)		ORIFICE ELEMENT
TI	DCS INSTRUMENT	TI	FIELD INSTRUMENT
TI	SIS FUNCTION		OFF-SHEET LINK
	CATALYST BED		S&T EXCHANGER

CONTROL LOOP SCHEDULE

LOOP	MEASURES	MOVES	FAIL	SET / NOTE
FIC-103	SULFUR TO F-101	FV-103	FC	40.9 t/h (PRODUCTION)
FFC-101	AIR/SULFUR RATIO	FIC-101 SP	-	4.99 kNm3/t (14 % SO2)
FIC-101	COMBUSTION AIR	FV-101	FO	CAS, MIN 100 kNm3/h
AIC-104	SO2 INTO BED 1	FIC-102 SP	-	12.0 % SO2
FIC-102	DILUTION AIR	FV-102	FO	CAS FROM AIC-104
TIC-201	BED 1 INLET	TV-201 (B-101 BYP.)	FC	420 C
TIC-211	BED 2 INLET	TV-211 (E-201 BYP.)	FO	445 C
TIC-221	BED 3 INLET	TV-221 (E-202 BYP.)	FO	440 C
TIC-231	BED 4 INLET	TV-231 (SH-201 BYP.)	FO	425 C (NOTE 4)
TIC-241	BED 5 INLET	TV-241 (E-204 BYP.)	FO	428 C
TIC-251	GAS TO T-301	TV-251 (E-203 BYP.)	FO	190 C
HIC-109	-	H-101 FIRING	FC	START-UP ONLY

SIS INTERLOCKS (ACTION: SULFUR CUT)

I-	INITIATOR	SETTING	ACTION
1	FSLL-100 LOW TOTAL AIR	< 90 kNm3/h, 3 s	CLOSE XV-103
2	TSHH-203 BED 1 EXIT	> 660 C, 5 s	CLOSE XV-103
3	FSLL-301 IPA ACID FLOW	< 50 %, 10 s	CLOSE XV-103
4	FSLL-302 FAT ACID FLOW	< 50 %, 10 s	CLOSE XV-103
5	ASHH-306 STACK SO2	> 3000 ppm, 5 min *	CLOSE XV-103
6	HS-001 ESD PUSHBUTTON	MANUAL	XV-103 + STOP K-101, H-101

DESIGN POINT (OTS MODEL)

BED	IN C	OUT C	X %	PDI kPa
1	420	631	63.1	4.5
2	445	528	88.1	3.0
3	440	465	95.8	3.6
4	425	441	99.70	2.9
5	428	427	99.80	3.0

12.0 % SO2 / 9.0 % O2 INTO BED 1; STACK 286 ppm SO2;
PI-200 35 kPa g; K-101 DISCHARGE 60 kPa g.

GENERAL NOTES:

1. DESIGN BASIS: 1000 kt/y H2SO4 (100 %) AT 8000 h/y = 125 t/h; SULFUR 40.9 t/h BURNED IN 204.5 kNm3/h DRY AIR (14 % SO2), DILUTED TO 12.0 % SO2 / 9.0 % O2, 238.5 kNm3/h. CONVERSION TARGETS 60/86/94/99.7/99.8 % PER PLANT SLIDE 5.2; BED TEMPERATURES FROM HEAT BALANCE (dH = -98.5 kJ/mol) - SEE DESIGN TABLE. GHSV 700 1/h, 340 m3 CATALYST (ASSUMED SPLIT 18/20/24/20/18 %).
2. DUCT AND LINE SIZES PRELIMINARY (VELOCITY 20-25 m/s GAS, 1.5 m/s SULFUR, 2.5 m/s ACID) - TBC BY PIPING. SERVICE CODES: PA PROCESS AIR, SG SO2 GAS, PG PROCESS GAS, MS MOLTEN SULFUR, SA SULFURIC ACID.
3. ALL CONTROL LOOPS IN DCS (SHARED DISPLAY). SIS FUNCTIONS I-1..I-6 IN A SEPARATE LOGIC SOLVER; XV-103 FAILS CLOSED. * ASHH-306 INHIBITED FOR 60 min AFTER SULFUR ADMISSION (START-UP).
4. BED 4 INLET IS SET BY THE HEAT LEFT IN THE BED-1 GAS FOR THE RETURN GAS: TIC-231 (TT-231 ON SO3-PID-002) BYPASSES SH-201; TIC-211 THEN HOLDS BED 2 VIA TV-211. LOOPS INTERACT - TUNE TIC-231 SLOW.
5. HV-112 START-UP HOT-GAS LINE LOCKED CLOSED IN OPERATION (ZSC-112 IS A PERMISSIVE FOR SULFUR ADMISSION); OPENED ONLY TO PREHEAT BEDS 4-5 BEFORE SULFUR IS ADMITTED.
6. H-101 PREHEAT BURNER: FLAME SAFEGUARD, FUEL SHUT-OFF VALVES AND IGNITER BY VENDOR; TRIPS ON LOW AIR (< 60 kNm3/h). START-UP ONLY.
7. B-101 STEAM SIDE (DRUM, LEVEL CONTROL, BLOWDOWN, PSV) AND SH-201 STEAM PIPING PER STEAM SYSTEM P&ID - OUTSIDE OTS SCOPE.

REV	DESCRIPTION	DATE
A	ISSUED FOR OTS DEVELOPMENT (REVIEW)	25-SEP-26

CHULA CHEM. ENG. - SO3 CONVERTER VER 1 (OTS)

P&ID - AIR, SULFUR FURNACE, CONVERTER BEDS 1-3

DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 25-SEP-26
DWG NO. SO3-PID-001	REV A	SCALE N.T.S.	SHEET 1 OF 2